

WEYL-CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL-RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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1. PRELIMINARIES

1.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\| \cdot \|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

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Define $d(x, y) = \min_{g \in G} |x - gy|$. Recall from [8, Chapter VII, proof of Theorem 2.12] that $d(x, y) = |x - y|$ iff x and y belong to the same chamber.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl}, j} := T_{e_j} (-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

1.2. Weyl chamber as simplicial cone.

Definition 1.1. A connected component of the set

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [2].

Theorem 1.2. Let Γ be a chamber. There exists a linear isomorphism $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$ (with $N_1 + N_2 = N$).

1.3. System of dyadic cubes.

Notation 1.3. *Let*

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube in $\mathbb{D}_k$ is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube $Q \in \mathbb{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

1.4. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [5, 9, 11, 3, 4, 10, 12, 14, 13]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [9, Section 13] for a complete proof).

Definition 1.4 and Lemma 1.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

Definition 1.4. *Let $w \in A_2(\mathbb{R}^N)$. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_Q\}_{Q \in \mathbb{D}}$ is called (r, c) -NWO if*

- (1) $\text{supp}(e_Q) \subseteq cQ$ for every $Q \in \mathbb{D}$;
- (2) $\|e_Q\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq w(Q)^{\frac{1}{r} - \frac{1}{2}}$ for every $Q \in \mathbb{D}$;

Lemma 1.5. *Let $w \in A_2(\mathbb{R}^N)$. Let $0 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, w(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, w(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)) \right) &\leq \\ &\leq C_{w,r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left(\{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [9]. □

2. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

Theorem 2.1. *Let Γ be a chamber. For all $x \in \Gamma$ and for all $r > 0$, we have*

$$w(B(x, r) \cap \Gamma) \geq c_\kappa w(B(x, r)).$$

In particular, $(\Gamma, |\cdot|, w(x)dx)$ is a doubling metric measure space.

Notation 2.2. Let $\alpha : 2^{N_1} \rightarrow \mathbb{R}_+$. Set

$$w_\alpha(x) = \prod_{A \subset \{1, \dots, N_1\}} \left(\sum_{k \in A} x_k \right)^{\alpha_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

Lemma 2.3. For every $\alpha : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have

$$\int_{B(x, r)} w_\alpha(y_1, \dots, y_{N_1}) dy \leq c_{N, \alpha}^{(1)} r^N w_\alpha((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \subset \left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} & \int_{B(x, r)} w_\alpha(y_1, \dots, y_{N_1}) dy \leq \\ & \leq \int_{\left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_\alpha(y_1, \dots, y_{N_1}) dy = \\ & = \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_\alpha(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ & \leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_\alpha((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\ & = w_\alpha((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ & \leq w_\alpha((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}. \end{aligned}$$

□

Lemma 2.4. For every $\alpha : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have

$$\int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\alpha(y_1, \dots, y_{N_1}) dy \geq c_{N, \alpha}^{(2)} r^N w_\alpha((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$x + \left[\frac{r}{2N}, \frac{r}{N} \right]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned} & \int_{B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\alpha(y_1, \dots, y_{N_1}) dy \geq \\ & \geq \int_{x + \left[\frac{r}{2N}, \frac{r}{N} \right]^N} w_\alpha(y_1, \dots, y_{N_1}) dy = \\ & = \int_{(x_1, \dots, x_{N_1}) + \left[\frac{r}{2N}, \frac{r}{N} \right]^{N_1}} w_\alpha(u) du \times \left(\frac{r}{2N} \right)^{N_2} \geq \\ & \geq \int_{(x_1, \dots, x_{N_1}) + \left[\frac{r}{2N}, \frac{r}{N} \right]^{N_1}} w_\alpha((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times \left(\frac{r}{2N} \right)^{N_2} = \\ & = w_\alpha((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times \left(\frac{r}{2N} \right)^{N_1} \geq \\ & \geq w_\alpha((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \left(\frac{r}{2N} \right)^N \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \alpha_A}. \end{aligned}$$

□

Lemma 2.5. *There exists $\alpha : 2^{N_1} \rightarrow \mathbb{R}_+$ such that*

$$c_\kappa^{-1} w_\alpha(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\alpha(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let A_v be the set of those $1 \leq k \leq N_1$ with $((\Phi^{-1})^*(v))_k$ being strictly positive. We, obviously, have

$$c_{\Phi, v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi, v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi, v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi, v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. $\square$

Proof of Theorem 2.1. The first assertion follows by combining Lemmas 2.3, 2.4 and 2.5. The second assertion is an obvious consequence of the first one. $\square$

3. FIRST ROCHBERG–SEMMES BASED ESTIMATE

Theorem 3.1. *Let $N > 1$. Fix a closed Weyl chamber Γ and $1_G \neq g \in G$. Suppose that $K : \Gamma \times \Gamma \rightarrow \mathbb{C}$ satisfies*

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{|x - gy| \cdot V(x, |x - y|)}, \quad x, y \in \Gamma, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator with integral kernel K . For every $f \in \dot{W}^{1, N}(\Gamma)$, we have

$$\|M_{f-f \circ g} V\|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \leq c_{K, \kappa} (\|f\|_{\dot{W}^{1, N}(\Gamma)} + \|f\|_{\dot{W}^{1, N}(g\Gamma)}).$$

Theorem 3.2. *In the setting of Theorem 3.1, let $f \in C_c^\infty$. If $0 \leq \phi_m \leq 1$ for $m \geq 1$ are in $C_c^\infty(\mathbb{R}^N)$ and $\text{supp}(1 - \phi_m)$ increases to $\mathbb{R}^N$, then*

$$\text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \left(M_{\phi_m} \cdot M_f V \cdot M_{\phi_m} - M_f V, (\mathcal{L}_{N, \infty})_0(L_2(\Gamma, w(x)dx)) \right) \rightarrow 0$$

as $m \rightarrow \infty$.

Notation 3.3. *Fix a closed Weyl chamber Γ and let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism given by Theorem 1.2. Let*

$$\mathbb{D}_\Gamma = \{ \Phi^{-1}(Q) \}_{\substack{Q \in \mathbb{D} \\ Q \subset \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}}}.$$

If $I \in \mathbb{D}_\Gamma$ and if $Q = \Phi(I)$, then we denote $d_I = \ell(Q)$.

3.1. Proof of Theorem 3.1.

Lemma 3.4. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that*

$$1 = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \chi_{3I}(y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$(3) \quad \text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(4) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every $x \in \Gamma$, there exists a unique $I \in \mathbb{D}_{\Gamma,j}$ containing x . For this I , we have $d_I = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(\Phi(x) - \Phi(y))) = \sum_{j \in \mathbb{Z}} \left(\sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \right) \psi(2^j(\Phi(x) - \Phi(y))) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right). \end{aligned}$$

If $x \in I$ and $\psi(\frac{\Phi(x) - \Phi(y)}{d_I}) \neq 0$, then

$$\Phi(x) \in \Phi(I) = Q, \quad |\Phi(x) - \Phi(y)| \leq d_I = \ell(Q).$$

We obviously have $|\Phi(x) - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|\Phi(x) - \Phi(y)|_\infty \leq \ell(Q)$. Thus, $|\Phi(y) - c_Q| \leq \frac{3}{2}\ell(Q)$. In other words, $\Phi(y) \in 3Q$ or, equivalently, $y \in 3I$. Thus,

$$\chi_I(x) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \chi_I(x) \chi_{3I}(y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right).$$

This completes the proof. $\square$

Lemma 3.5. *Assume that $N > 1$ and $2 < r < \infty$. Let $f \in L_r^{\text{loc}}(\mathbb{R}^N)$. Fix a closed Weyl chamber Γ and $1_G \neq g \in G$. Let K be as in Theorem 3.1. We have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k, l, I}(x) f_{k, l, I}(y),$$

where

- (1) $e_{k, l, I}$ is supported in I and $f_{k, l, I}$ is supported in $3I$;
- (2) for every $k, l \in \mathbb{Z}^N$ and for every $I \in \mathbb{D}_\Gamma$, we have

$$\|e_{k, l, I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = \|f\|_{L_r(I, w(x)dx)};$$

- (3) for every $k, l \in \mathbb{Z}^N$ and for every $I \in \mathbb{D}_\Gamma$, we have

$$\|f_{k, l, I}\|_{L_\infty(\mathbb{R}^N)} \leq c_{K, \kappa, \Phi} \frac{d_I}{|c_I - gc_I|};$$

Proof. We have

$$\begin{aligned} K(x, y) &= \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) = \\ &= \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right) + \sum_{\substack{I \in \mathbb{D}_\Gamma \\ 3I \not\subset \Gamma}} \chi_I(x) \chi_{3I}(y) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{d_I}\right). \end{aligned}$$

We will only consider the first sum. The second one can be treated similarly, but the notations become heavier. **feel free to write the expression for the second sum if you think it is needed. the only difference is that the domain of the map K_I becomes more complicated.**

For every $I \in \mathbb{D}_\Gamma$ with $3I \subset \Gamma$, define the map

$$K_I : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_I(a, b) = K(c_I + d_I \Phi^{-1}(a), c_I + d_I \Phi^{-1}(b)) \psi(a - b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

It follows from the assumption on K that **feel free to prove this if you think this requires a proof. however, do not prove it here — make a separate lemma instead.**

$$\|K_I\|_{C^{2N+2}([-1/2, 1/2]^N \times [-3/2, 3/2]^N)} \leq \frac{c'_{K, \kappa, \Phi}}{w(I)} \cdot \frac{d_I}{|c_I - gc_I|}.$$

Hence, there exists an extension $K_I : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_I\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K, \kappa, \Phi}}{w(I)} \cdot \frac{d_I}{|c_I - gc_I|}.$$

In particular,

$$\|K_I\|_{W^{2N+2, 2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K, \kappa, \Phi}}{w(I)} \cdot \frac{d_I}{|c_I - gc_I|}.$$

We may, therefore, write

$$K_I(a, b) = \sum_{k, l \in \mathbb{Z}^N} \text{coef}_{k, l, I} e^{i\langle k, a \rangle} e^{i\langle l, b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k, l, I}|^2 \leq \left(\frac{c_{K, \kappa, \Phi}}{w(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{-N-1} \cdot |\text{coef}_{k, l, I}| \leq \frac{c_{K, \kappa, \Phi}}{w(I)} \cdot \frac{d_I}{|c_I - gc_I|}, \quad k, l \in \mathbb{Z}^N, \quad I \in \mathbb{D}_\Gamma.$$

We now set

$$e_{k, l, I}(x) = f(x) \chi_I(x) e^{i\langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle},$$

$$f_{k, l, I}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k, l, I} \cdot \chi_{3I}(y) e^{i\langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle}.$$

This allows us to write

$$f(x) K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k, l, I}(x) f_{k, l, I}(y).$$

Obviously,

$$\begin{aligned} \text{supp}(e_{k,l,I}) &\subset I, \quad \text{supp}(f_{k,l,I}) \subset 3I, \\ \|e_{k,l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= \|f\|_{L_r(I, w(x)dx)}, \\ \|f_{k,l,I}\|_{L_\infty(\mathbb{R}^N, w(x)dx)} &= (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{I,k,l}| \leq \frac{c_{K,\kappa,\Phi}}{w(I)} \cdot \frac{d_I}{|c_I - gc_I|}. \end{aligned}$$

□

Definition 3.6. Let Γ be a chamber. Let Φ be as in Theorem 1.2 and let $\mathbb{D}_\Gamma$ be as in Notation 3.3. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_I\}_{I \in \mathbb{D}_\Gamma}$ is called (r, c) -NWO if

- (1) $\text{supp}(e_I) \subseteq cI$ for every $I \in \mathbb{D}_\Gamma$;
- (2) $\|e_I\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq w(I)^{\frac{1}{r} - \frac{1}{2}}$ for every $I \in \mathbb{D}_\Gamma$;

Lemma 3.7. Let $0 < p < \infty$. Let Γ be a chamber. Let Φ be as in Theorem 1.2 and let $\mathbb{D}_\Gamma$ be as in Notation 3.3. Suppose that A is an integral operator on $L_2(\Gamma, w(x)dx)$ with integral kernel

$$(x, y) \rightarrow \sum_{I \in \mathbb{D}_\Gamma} \lambda_I e_I(x) f_I(y), \quad x, y \in \Gamma,$$

where $\{e_I\}_{I \in \mathbb{D}_\Gamma}$ and $\{f_I\}_{I \in \mathbb{D}_\Gamma}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\Gamma, w(x)dx))} &\leq C_{\Phi,\kappa,r_1,c_1,r_2,c_2,p} \|\{\lambda_I\}_{I \in \mathbb{D}_\Gamma}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, w(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\Gamma, w(x)dx)) \right) &\leq \\ &\leq C_{\Phi,\kappa,r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D}_\Gamma)} \left(\{\lambda_I\}_{I \in \mathbb{D}_\Gamma}, (l_{p,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

Proof. The assertion follows immediately from Lemma 1.5. □

Lemma 3.8. For every $I \in \mathbb{D}_\Gamma$, we have

$$\sup_{x \in I} w(x) \leq c_{\kappa,\Phi} \frac{w(I)}{m(I)}.$$

Proof. We have

$$w(I) \approx_{\Phi,\kappa} w(B(c_I, d_I)) \approx_\kappa d_I^N \prod_{v \in R} (|\langle c_I, v \rangle| + d_I)^{k(v)}.$$

On the other hand, if $x \in I$, then $|x - c_I| \leq c_\Phi d_I$ and, therefore,

$$|\langle x, v \rangle| \leq |\langle x - c_I, v \rangle| + |\langle c_I, v \rangle| \leq |x - c_I| + |\langle c_I, v \rangle| \leq c_\Phi d_I + |\langle c_I, v \rangle|.$$

Thus,

$$w(x) \leq \prod_{v \in R} (c_\Phi d_I + |\langle c_I, v \rangle|)^{k(v)} \leq c_{\Phi,\kappa} \prod_{v \in R} (|\langle c_I, v \rangle| + d_I)^{k(v)}.$$

This completes the proof. □

Lemma 3.9. Assume that $N > 1$ and $2 < r < \infty$. Let $f \in L_r^{\text{loc}}(\mathbb{R}^N)$. We have

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \leq c_{K,\kappa,\Phi,r} \left\| \left\{ m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty}.$$

Proof. Let $A_{k,l}$ be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{I \in \mathbb{D}_\Gamma} e_{k,l,I}(x) f_{k,l,I}(y), \quad x, y \in \Gamma, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 3.5, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \leq \\ & \leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))}. \end{aligned}$$

By Lemma 3.5, the sequences

$$\left\{ w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,I}}{\|f\|_{L_r(I, w(x)dx)}} \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ c_{K,\kappa,\Phi}^{-1} w(I)^{\frac{1}{2}} \cdot \frac{|c_I - gc_I|}{d_I} f_{k,l,I} \right\}_{I \in \mathbb{D}_\Gamma}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 3.6. Using Lemma 3.7, we conclude that

$$\begin{aligned} & \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \leq \\ & \leq c_{\Phi,\kappa,r,1,\infty,3,N} c_{K,\kappa} \left\| \left\{ w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \leq \\ & \leq \frac{N}{N-1} c_{\Phi,\kappa,r,1,\infty,3,N} c_{K,\kappa,\Phi} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \\ & \times \left\| \left\{ w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty}. \end{aligned}$$

By Lemma 3.8,

$$w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \leq c_{\kappa,\Phi}^{\frac{1}{r}} m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)}, \quad I \in \mathbb{D}_\Gamma.$$

Combining the last two estimates, we complete the proof. $\square$

Lemma 3.10. *Let $f \in \dot{W}^{1,N}(\Gamma) \cap \dot{W}^{1,N}(g\Gamma)$. We have*

$$\left\| \left\{ m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} \leq c_\Phi (\|f\|_{\dot{W}^{1,N}(\Gamma)} + \|f\|_{\dot{W}^{1,N}(g\Gamma)}).$$

Proof. Let $\mathcal{M}$ be the Hardy–Littlewood maximal operator on $\mathbb{R}^N$. By [?, Theorem 3], for almost every $x \in \mathbb{R}^N$,

$$|f(x) - f(gx)| \leq c_N |x - gx| \cdot ((\mathcal{M}(|\nabla f|))(x) + (\mathcal{M}(|\nabla f|))(gx)).$$

We have

$$|x - gx| \leq 2|x - c_I| + |c_I - gc_I|, \quad x \in I, \quad y \in gI.$$

Obviously,

$$|x - c_I| \leq \frac{d_I^{\frac{1}{2}}}{2} \|\Phi\|_\infty d_I, \quad x \in I.$$

Hence,

$$|x - gx| \leq c'_\Phi |c_I - gc_I|, \quad x \in I.$$

Thus,

$$|f(x) - f(gx)| \leq c_\Phi'' \cdot |c_I - gc_I| \cdot ((\mathcal{M}(|\nabla f|))(x) + (\mathcal{M}(|\nabla f|))(gx)).$$

This immediately implies

$$\frac{1}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \leq c_\Phi'' \cdot \left(\|\mathcal{M}(|\nabla f|)\|_{L_1(I)} + \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right)$$

for every $I \in \mathbb{D}_\Gamma$. Hence,

$$\begin{aligned} m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{|c_I - gc_I|} &\leq \\ &\leq c_\Phi''' \cdot m(I)^{\frac{1}{N}-1} \left(\|\mathcal{M}(|\nabla f|)\|_{L_1(I)} + \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right) \end{aligned}$$

for every $I \in \mathbb{D}_\Gamma$. Thus,

$$\begin{aligned} &\left\| \left\{ m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} \leq \\ &\leq c_\Phi''' \frac{N}{N-1} \left(\left\| \left\{ m(I)^{\frac{1}{N}-1} \|\mathcal{M}(|\nabla f|)\|_{L_1(I)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} + \right. \\ &\quad \left. + \left\| \left\{ m(I)^{\frac{1}{N}-1} \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} \right) \leq \\ &\stackrel{L.A.1}{\leq} c_\Phi''' \frac{N}{N-1} c_N \left(\|\mathcal{M}(|\nabla f|)\|_{L_N(\Gamma)} + \|\mathcal{M}(|\nabla f|)\|_{L_N(g\Gamma)} \right) \leq \\ &\leq c_\Phi''' \frac{N}{N-1} c_N \|\mathcal{M}\|_{L_N(\mathbb{R}^N) \odot} \left(\|\nabla f\|_{L_N(\Gamma)} + \|\nabla f\|_{L_N(g\Gamma)} \right). \end{aligned}$$

□

Lemma 3.11. *Let $r > N$ and let $f \in \dot{W}^{1,N}(\Gamma) \cap \dot{W}^{1,N}(g\Gamma)$. We have*

$$\left\| \left\{ m(I)^{-\frac{1}{r}} \|f - f \circ g\|_{L_r(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} \leq c_\Phi (\|f\|_{\dot{W}^{1,N}(\Gamma)} + \|f\|_{\dot{W}^{1,N}(g\Gamma)}).$$

Proof. We obviously have

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \left\| f - \frac{1}{m(I)} \int_I f \right\|_{L_r(I)} + \\ &+ \left\| f \circ g - \frac{1}{m(I)} \int_I (f \circ g) \right\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)}. \end{aligned}$$

Let $p \in (0, N)$ be such that $\frac{1}{r} = \frac{1}{p} - \frac{1}{N}$. By Sobolev inequality,

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \\ &\leq c'_{\Phi,r} \left(\|\nabla f\|_{L_p(I)} + \|\nabla(f \circ g)\|_{L_p(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)} \right). \end{aligned}$$

By the triangle inequality,

$$\begin{aligned} &\left\| \left\{ m(I)^{-\frac{1}{r}} \|f - f \circ g\|_{L_r(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} \leq \\ &\leq c''_{\Phi,r} \left\| \left\{ m(I)^{\frac{1}{N}-\frac{1}{p}} \|\nabla f\|_{L_p(I)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} + \\ &+ c''_{\Phi,r} \left\| \left\{ m(I)^{\frac{1}{N}-\frac{1}{p}} \|\nabla f\|_{L_p(gI)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty} + \\ &+ c''_{\Phi,r} \left\| \left\{ m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N,\infty}. \end{aligned}$$

The assertion follows now from Lemma A.1 (for the first two summands) and Lemma 3.10 (for the last summand). $\square$

Proof of Theorem 3.1. The assertion follows by combining Lemmas 3.9 (taken with $f - f \circ g$ instead of f) and 3.11. $\square$

3.2. Proof of Theorem 3.2.

Lemma 3.12. *Let $f, \phi \in C_c^\infty(\mathbb{R}^N)$. If $0 \leq \phi \leq 1$, then*

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left(M_f V - M_{f\phi} V, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq c_{K,\kappa,\phi,r} \times \\ & \quad \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right), \\ & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left(M_f V - M_f V M_\phi, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq c_{K,\kappa,\phi,r} \times \\ & \quad \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

Proof. We only prove the second inequality. The proof of the first one is similar, but simpler.

Let $A_{k,l}$ be as in the proof of Lemma 3.9. We have

$$M_f V - M_f V M_\phi = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l} M_{1-\phi}.$$

By the triangle inequality,

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left(M_f V - M_f V M_\phi, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} \left((1 + |k|^2 + |l|^2)^{-N-1} \times \right. \\ & \quad \left. \times \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left(A_{k,l} M_{1-\phi}, (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \right). \end{aligned}$$

By Lemma 3.5, the sequences

$$\left\{ w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,I}}{\|f\|_{L_r(I, w(x)dx)}} \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ c_{K,\kappa,\Phi}^{-1} w(I)^{\frac{1}{2}} \cdot \frac{|c_I - gc_I|}{d_I} f_{k,l,I} \cdot (1-\phi) \right\}_{I \in \mathbb{D}_\Gamma}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 3.6. Obviously, $e_{k,l,I} \not\equiv 0$ only if $I \cap \text{supp}(f) \neq \emptyset$ and $f_{k,l,I} \cdot (1-\phi) \not\equiv 0$ only if $I \cap \text{supp}(f) \neq \emptyset$ and $3I \cap \text{supp}(1-\phi) \neq \emptyset$. Using Lemma 3.7, we conclude that

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left(A_{k,l}(1-\phi), (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq c_{\Phi,\kappa,r,1,\infty,3,N} c_{K,\kappa,\Phi} \times \\ & \quad \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I w(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f\|_{L_r(I, w(x)dx)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

By Lemma 3.8,

$$w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \leq c_{\kappa,\Phi}^{\frac{1}{r}} m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)}, \quad I \in \mathbb{D}_\Gamma.$$

Thus,

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left(A_{k,l}(1-\phi), (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq c_{\Phi, \kappa, r, 1, \infty, 3, N} c_{K, \kappa, \Phi} C_{\kappa, \Phi}^{\frac{1}{r}} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

Combining this estimate with the one in the first paragraph [assign label and refer to it](#) yields the second inequality. $\square$

Lemma 3.13. *For a bounded Borel $A \subset \mathbb{R}^N$, we have*

$$\lim_{n \rightarrow \infty} \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset A \\ \ell(Q) \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq c_N m(A)^{\frac{1}{N}}.$$

Proof. For a given $\epsilon > 0$, choose j (depending on ϵ) and finite set $B \subset \mathbb{D}_j$ such that $A \subset \cup_{Q' \in B} Q'$ and such that $m((\cup_{Q' \in B} Q') \setminus A) < \epsilon$. For $n \geq m$, we have (here, we use the notation $Q_0 = [0, 1)^N$)

$$\begin{aligned} & \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset A \\ \ell(Q) \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset Q' \text{ for some } Q' \in B}} \right\|_{l_{N,\infty}(\mathbb{D})} = \\ & = \left(\frac{|B|}{2^j} \right)^{\frac{1}{N}} \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset Q_0}} \right\|_{l_{N,\infty}(\mathbb{D})} = c_N \cdot m^{\frac{1}{N}}(\cup_{Q' \in B} Q') \leq c_N \cdot (m(A) + \epsilon)^{\frac{1}{N}}. \end{aligned}$$

Passing $n \rightarrow \infty$ and, after that, passing $\epsilon \rightarrow 0$, we complete the proof. $\square$

Lemma 3.14. *We have*

$$\lim_{n \rightarrow \infty} \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq c_N m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1-\phi)).$$

Proof. Denote for brevity $A = \Phi^{-1}(\text{supp}(f))$ and $B = \Phi^{-1}(\text{supp}(1-\phi))$. For $\epsilon > 0$, denote by A_ϵ (respectively, B_ϵ) the ϵ -neighborhood of A (respectively, of B) with respect to the norm $|\cdot|_\infty$.

If $2^{-n} < \epsilon$, then the condition $I \cap \text{supp}(\phi) \neq \emptyset$ yields, for $Q = \Phi(I) \in \mathbb{D}$, the condition $Q \subset A_\epsilon$. If $2^{-n} < \frac{\epsilon}{3}$, then the condition $I \cap \text{supp}(1-\phi) \neq \emptyset$ yields, for $Q = \Phi(I) \in \mathbb{D}$, the condition $Q \subset B_\epsilon$. Hence, for $2^{-n} < \frac{\epsilon}{3}$, we have

$$\left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq \det(\Phi)^{-\frac{1}{N}} \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset A_\epsilon \cap B_\epsilon \\ \ell(Q) \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D})}.$$

Using Lemma 3.13 and passing $n \rightarrow \infty$, we obtain

$$\lim_{n \rightarrow \infty} \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq c_N \det(\Phi)^{-\frac{1}{N}} m^{\frac{1}{N}}(A_\epsilon \cap B_\epsilon).$$

Since A and B are both closed, it follows that

$$\lim_{\epsilon \rightarrow 0} m(A_\epsilon \cap B_\epsilon) = m(A \cap B).$$

Passing $\epsilon \rightarrow 0$ in the second last display, we complete the proof. $\square$

Lemma 3.15. *Let $r > 2$. For every $f \in C_c^\infty(\mathbb{R}^N)$ and for every $n \in \mathbb{N}$, we have*

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq c_N \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1-\phi)). \end{aligned}$$

Proof. For a given $n \in \mathbb{Z}$, we have

$$\left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}} \leq \|\nabla f\|_{L_r(\mathbb{R}^N)} \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}}.$$

The right hand side obviously belongs to $(l_{N,\infty})_0(\mathbb{D}_\Gamma)$ and, hence, so does the left hand side. Thus,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq \left\| \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq \\ \leq \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Passing $n \rightarrow \infty$ and using Lemma 3.14, we complete the proof. $\square$

Lemma 3.16. *For every $f \in C_c^\infty(\mathbb{R}^N)$ and for every $n \in \mathbb{N}$, we have*

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-1}}{|c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1-\phi)). \end{aligned}$$

Proof. For a given $n \in \mathbb{Z}$, we have

$$\left\{ \frac{d_I m(I)^{-1}}{|c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}} \leq c'_\Phi \|f\|_{L_1(\mathbb{R}^N)} \left\{ m(I)^{-1} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}.$$

The right hand side obviously belongs to $(l_{N,\infty})_0(\mathbb{D}_\Gamma)$ and, hence, so does the left hand side. Thus,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-1}}{|c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ \leq \left\| \left\{ \frac{d_I m(I)^{-1}}{|c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Obviously,

$$|f(x) - f(y)| \leq c_N |x - y| \cdot \|\nabla f\|_{L_\infty(\mathbb{R}^N)}.$$

Hence,

$$\begin{aligned} \|f - f \circ g\|_{L_1(I)} &\leq c_N \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \int_I |x - gx| dx \leq \\ &\leq c_\Phi'' m(I) \times |c_I - gc_I| \times \|\nabla f\|_{L_\infty(\mathbb{R}^N)}. \end{aligned}$$

Thus,

$$\frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \leq c_\Phi''' m(I)^{\frac{1}{N}} \|\nabla f\|_{L_\infty(\mathbb{R}^N)}$$

Finally,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) &\leq \\ &\leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Passing $n \rightarrow \infty$ and using Lemma 3.14, we complete the proof. $\square$

Lemma 3.17. *Let $r > 2$. For every $f \in C_c^\infty(\mathbb{R}^N)$ and for every $n \in \mathbb{N}$, we have*

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f - f \circ g\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) &\leq \\ &\leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1-\phi)). \end{aligned}$$

Proof. We obviously have

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \left\| f - \frac{1}{m(I)} \int_I f \right\|_{L_r(I)} + \\ &+ \left\| f \circ g - \frac{1}{m(I)} \int_I (f \circ g) \right\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)}. \end{aligned}$$

By Poincare inequality,

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} &\leq \\ &\leq c'_{\Phi,r} \left(m(I)^{\frac{1}{N}} \|\nabla f\|_{L_r(I)} + m(I)^{\frac{1}{N}} \|\nabla(f \circ g)\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)} \right). \end{aligned}$$

By the triangle inequality,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f - f \circ g\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) &\leq \\ &\leq c''_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) + \\ &+ c''_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(gI)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) + \\ &+ c''_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left(\left\{ \frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

The assertion follows from Lemmas 3.15 and 3.16. $\square$

Proof of Theorem 3.2. The assertion follows by combining Lemmas 3.12 (applied with $f - f \circ g$ instead of f) and 3.17. $\square$

4. SECOND ROCHBERG-SEMMES BASED ESTIMATE

Theorem 4.1. *Let $N > 1$. Fix a closed Weyl chamber Γ . Suppose that $K : \Gamma \times \Gamma \rightarrow \mathbb{C}$ satisfies*

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{-|\alpha|_1 - |\beta|_1}}{V(x, |x - y|)}, \quad x, y \in \Gamma, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator with integral kernel K . For every $f \in \dot{W}^{1, N}(\Gamma)$, we have

$$\| [M_f, V] \|_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \leq c_{K, \kappa} (\|f\|_{\dot{W}^{1, N}(\Gamma)} + \|f\|_{\dot{W}^{1, N}(g\Gamma)}).$$

Theorem 4.2. *In the setting of Theorem 3.1, let $f \in C_c^\infty$. If $0 \leq \phi_m \leq 1$ for $m \geq 1$ are in $C_c^\infty(\mathbb{R}^N)$ and $\text{supp}(1 - \phi_m)$ increases to $\mathbb{R}^N$, then*

$$\text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \left(M_{\phi_m} \cdot [M_f, V] \cdot M_{\phi_m} - [M_f, V], (\mathcal{L}_{N, \infty})_0(L_2(\Gamma, w(x)dx)) \right) \rightarrow 0$$

as $m \rightarrow \infty$.

Lemma 4.3. *Assume that $N > 1$ and $2 < r < \infty$. Let $f \in L_r^{\text{loc}}(\mathbb{R}^N)$. Fix a closed Weyl chamber Γ . Let K be as in Theorem 3.1. We have*

$$\begin{aligned} (f(x) - f(y)) \cdot K(x, y) &= \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k, l, I}^{(1)}(x) f_{k, l, I}^{(1)}(y) - \\ &\quad - \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k, l, I}^{(2)}(x) f_{k, l, I}^{(2)}(y) \end{aligned}$$

where

- (1) $e_{k, l, I}^{(1)}$ and $e_{k, l, I}^{(2)}$ are supported in I and $f_{k, l, I}^{(1)}$ and $f_{k, l, I}^{(2)}$ are supported in $3I$;
- (2) for every $k, l \in \mathbb{Z}^N$ and for every $I \in \mathbb{D}_\Gamma$, we have

$$\begin{aligned} \|e_{k, l, I}^{(1)}\|_{L_r(\mathbb{R}^N, w(x)dx)} &\leq \left\| f - \frac{1}{m(3I)} \int_{3I} f \right\|_{L_r(3I, w(x)dx)}; \\ \|f_{k, l, I}^{(2)}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= \left\| f - \frac{1}{m(3I)} \int_{3I} f \right\|_{L_r(3I, w(x)dx)}; \end{aligned}$$

- (3) for every $k, l \in \mathbb{Z}^N$ and for every $I \in \mathbb{D}_\Gamma$, we have

$$\|f_{k, l, I}^{(1)}\|_{L_\infty(\mathbb{R}^N)}, \|e_{k, l, I}^{(2)}\|_{L_\infty(\mathbb{R}^N)} \leq c_{K, \kappa, \Phi};$$

Proof. Repeating the argument in Lemma 3.5 *mutatis mutandi*, we obtain

$$(1 + |k|^2 + |l|^2)^{-N-1} \cdot |\text{coef}_{k, l, I}| \leq \frac{c_{K, \kappa, \Phi}}{w(I)}, \quad k, l \in \mathbb{Z}^N, \quad I \in \mathbb{D}_\Gamma.$$

Set

$$\begin{aligned} e_{k, l, I}^{(1)}(x) &= \left(f(x) - \frac{1}{m(3I)} \int_{3I} f \right) \cdot \chi_I(x) e^{i \langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle}, \\ f_{k, l, I}^{(1)}(y) &= (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k, l, I} \cdot \chi_{3I}(y) e^{i \langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle}, \\ e_{k, l, I}^{(2)}(x) &= (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k, l, I} \cdot \chi_I(x) e^{i \langle k, \frac{\Phi^{-1}(x - c_I)}{d_I} \rangle}, \\ f_{k, l, I}^{(2)}(y) &= \left(f(y) - \frac{1}{m(3I)} \int_{3I} f \right) \cdot \chi_{3I}(y) e^{i \langle l, \frac{\Phi^{-1}(y - c_I)}{d_I} \rangle}. \end{aligned}$$

The assertion follows immediately. $\square$

to be written by Zhijie

5. UPPER ESTIMATE

The following lemma establishes derivative estimates for the Dunkl–Riesz transform kernel.

Lemma 5.1. *Assume that $N \geq 1$. Then for every pair of multi-indices α, β , every $j \in \{1, \dots, N\}$, and every $x, y \in \mathbb{R}^N$ with $d(x, y) > 0$,*

$$(5) \quad \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

Proof. **Case 1.** Suppose that $|\alpha| + |\beta| = 0$. By [6, Theorem 1.1],

$$|K_{\text{Dunkl},j}(x, y)| \lesssim \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

By the definition of d , the G -invariance of dw , and (2),

$$V(x, y, d(x, y)) \lesssim w(B(x, d(x, y))),$$

and hence (5) follows.

Case 2. Suppose that $|\alpha| + |\beta| \geq 1$. By [?, Theorem 3.1],

$$\left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \lesssim_{\alpha, \beta, R, k} t^{-\frac{|\alpha|+|\beta|}{2}} \left(1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t}.$$

Recall from the proof of Lemma ?? that

$$K_{\text{Dunkl},j}(x, y) = -\frac{C}{2} \int_0^\infty (y_j - x_j) h_t(x, y) \frac{dt}{t^{3/2}}.$$

Since

$$\partial_x^\alpha \partial_y^\beta ((y_j - x_j) h_t(x, y)) = (y_j - x_j) \partial_x^\alpha \partial_y^\beta h_t(x, y) - \alpha_j \partial_x^{\alpha - e_j} \partial_y^\beta h_t(x, y) + \beta_j \partial_x^\alpha \partial_y^{\beta - e_j} h_t(x, y),$$

where the corresponding term is omitted if $\alpha_j = 0$ or $\beta_j = 0$, we obtain

$$\begin{aligned} \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y) \right| &\lesssim \frac{1}{\|x - y\|} \int_0^\infty t^{-\frac{|\alpha|+|\beta|+1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\ &\quad + \frac{1}{\|x - y\|^2} \int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt. \end{aligned}$$

By (2),

$$V(x, y, \sqrt{t})^{-1} \lesssim V(x, y, d(x, y))^{-1} \begin{cases} \left(\frac{d(x, y)}{\sqrt{t}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{\sqrt{t}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Hence,

$$\begin{aligned}
& \int_0^\infty t^{-\frac{|\alpha|+|\beta|+1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\
& \lesssim \frac{1}{V(x, y, d(x, y))} \left(d(x, y)^N \int_0^{d(x, y)^2} t^{-\frac{N+|\alpha|+|\beta|+1}{2}} e^{-cd(x, y)^2/t} dt + d(x, y)^N \int_{d(x, y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|+1}{2}} dt \right) \\
& \lesssim \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))}.
\end{aligned}$$

The first integral in the parentheses is estimated by the change of variables $s = d(x, y)^2/t$, while the second one converges since $N + |\alpha| + |\beta| > 1$.

Case 2.1. Suppose that $N + |\alpha| + |\beta| > 2$. We apply the same argument to see that

$$\begin{aligned}
& \int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\
& \lesssim \frac{1}{V(x, y, d(x, y))} \left(d(x, y)^N \int_0^{d(x, y)^2} t^{-\frac{N+|\alpha|+|\beta|}{2}} e^{-cd(x, y)^2/t} dt + d(x, y)^N \int_{d(x, y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|}{2}} dt \right) \\
& \lesssim \frac{d(x, y)^{2-|\alpha|-|\beta|}}{V(x, y, d(x, y))}.
\end{aligned}$$

Since $d(x, y) \leq \|x - y\|$, it follows that

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

Case 2.2. Suppose that $N = 1$ and $|\alpha| + |\beta| = 1$. For the terms in which a derivative falls on $y_j - x_j$, retaining the factor $(1 + \|x - y\|/\sqrt{t})^{-2}$ and splitting the integral at $d(x, y)^2$ and $\|x - y\|^2$, we obtain from (2)

$$\begin{aligned}
& \int_0^\infty t^{-3/2} \left(1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\
& \lesssim \frac{d(x, y)^N}{\|x - y\|^2 V(x, y, d(x, y))} \int_0^{d(x, y)^2} t^{-\frac{N+1}{2}} e^{-cd(x, y)^2/t} dt \\
& \quad + \frac{d(x, y)}{\|x - y\|^2 V(x, y, d(x, y))} \int_{d(x, y)^2}^{\|x - y\|^2} \frac{dt}{t} + \frac{d(x, y)}{V(x, y, d(x, y))} \int_{\|x - y\|^2}^\infty \frac{dt}{t^2} \\
& \lesssim \frac{1}{\|x - y\| V(x, y, d(x, y))},
\end{aligned}$$

where we used

$$\frac{d(x, y)}{\|x - y\|} \log \frac{\|x - y\|}{d(x, y)} \lesssim 1.$$

Combining this estimate with the preceding estimate for the first term, we obtain

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{1}{\|x - y\| V(x, y, d(x, y))} = \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-1}}{V(x, y, d(x, y))}.$$

This completes the proof of Lemma 5.1. $\square$

6. SPECTRAL ASYMPTOTIC FORMULA

Definition 6.1. For $g \in G$ and $j = 1, \dots, N$, set

$$d_{g,j} := g^{-1}e_j, \quad f_g(x) := f(gx).$$

For $\alpha \in R_+$, define $S_\alpha : \mathbb{C}^G \rightarrow \mathbb{C}^G$ by

$$(S_\alpha z)_g := z_{g\sigma_\alpha}, \quad g \in G.$$

For $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, $x \in \Gamma_+$, and $\theta \in \mathbb{S}^{N-1}$, define

$$\begin{aligned} \mathbf{c}_{f,j}(x, \theta) &:= \text{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &\quad + \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

For smooth f , the quotients in the second term are understood by their smooth extensions across the corresponding reflection hyperplanes. For $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, $\mathbf{c}_{f,j}$ is understood almost everywhere on $\Gamma \times \mathbb{S}^{N-1}$.

Lemma 6.2. Let $f \in W_{\text{loc}}^{1,1}(\mathbb{R}^N)$. For every $u \in C_c^\infty(\mathbb{R}^N)$,

$$(1) \quad [T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x)$$

for almost every $x \in \mathbb{R}^N$. If $N > 1$ and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then, for every $\alpha \in R_+$,

$$(2) \quad \left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. Pairing α and $-\alpha$ in the definition of T_{e_j} , and using $k(-\alpha) = k(\alpha)$ and $\sigma_{-\alpha} = \sigma_\alpha$, we have

$$T_{e_j}u(x) = \partial_j u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{u(x) - u(\sigma_\alpha x)}{\langle \alpha, x \rangle}.$$

Therefore,

$$[T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x),$$

which proves (1).

Since R is normalized,

$$x - \sigma_\alpha x = \langle \alpha, x \rangle \alpha, \quad \|x - \sigma_\alpha x\| = \sqrt{2} |\langle \alpha, x \rangle|.$$

Hence, for almost every $x \notin \alpha^\perp$,

$$\frac{|f(x) - f(\sigma_\alpha x)|}{|\langle \alpha, x \rangle|} = \sqrt{2} \mathcal{G}_{I, \sigma_\alpha} f(x).$$

By Lemma ?? (??),

$$\left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

This completes the proof of Lemma 6.2. $\square$

Lemma 6.3. *Let Γ_+ be an open Weyl chamber, let Γ be its closure, and let R_+ be the corresponding positive subsystem. Set*

$$\rho(x) := \varpi(x)^{1/2},$$

and define

$$\begin{aligned} U : L^2(\mathbb{R}^N, dw) &\longrightarrow L^2(\Gamma, dw; \mathbb{C}^G), & (Uu)_g(x) &:= u(gx), \\ V : L^2(\Gamma, dw; \mathbb{C}^G) &\longrightarrow L^2(\Gamma, dx; \mathbb{C}^G), & (Vh)(x) &:= \rho(x)h(x). \end{aligned}$$

Then U and V are unitary. If

$$\tilde{T}_j := VUT_{e_j}U^{-1}V^{-1}, \quad \tilde{P} := VU(-\Delta_{\text{Dunkl}})U^{-1}V^{-1},$$

then, on $C_c^\infty(\Gamma_+; \mathbb{C}^G)$,

$$(3) \quad (\tilde{T}_j h)_g = \partial_{d_{g,j}} h_g - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} h_{g\sigma_\alpha},$$

$$(4) \quad \tilde{P} = -\Delta_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Assume in addition that $N > 1$, and let

$$\tilde{Q} := VU(-\Delta_{\text{Dunkl}})^{-1/2}U^{-1}V^{-1},$$

where $(-\Delta_{\text{Dunkl}})^{-1/2}$ is defined by the spectral calculus. Then, for every $\chi \in C_c^\infty(\Gamma_+)$, the operator $M_\chi \tilde{Q} M_\chi$, initially defined on its natural domain, admits a bounded extension to $L^2(\Gamma, dx; \mathbb{C}^G)$, and this extension is a classical matrix-valued pseudodifferential operator of order -1 . Its principal symbol is

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

Consequently, if $x \in \Gamma_+$, $\xi \neq 0$, and $\chi \in C_c^\infty(\Gamma_+)$ is identically 1 in a neighborhood of x , then

$$(5) \quad \sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}.$$

Proof. The Weyl chambers $g\Gamma_+$, $g \in G$, are pairwise disjoint and cover $\mathbb{R}^N$ up to the reflection hyperplanes; see [7, Theorem 2.12]. Since dw is G -invariant,

$$\begin{aligned} \|Uu\|_{L^2(\Gamma, dw; \mathbb{C}^G)}^2 &= \sum_{g \in G} \int_{\Gamma} |u(gx)|^2 dw(x) \\ &= \int_{\mathbb{R}^N} |u(x)|^2 dw(x). \end{aligned}$$

Conversely, for $h = (h_g)_{g \in G} \in L^2(\Gamma, dw; \mathbb{C}^G)$, define $u(gx) := h_g(x)$ for $x \in \Gamma_+$. This determines u almost everywhere on $\mathbb{R}^N$ and gives $Uu = h$. Hence U is unitary. Since $dw = \rho^2 dx$ and $\rho > 0$ on Γ_+ , V is unitary as well.

Pairing the roots α and $-\alpha$ in the definition of T_{e_j} , changing variables in the root system by g , and using the G -invariance of k , we obtain, for $h \in C_c^\infty(\Gamma_+; \mathbb{C}^G)$,

$$(UT_{e_j}U^{-1}h)_g = \partial_{d_{g,j}} h_g + \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (h_g - h_{g\sigma_\alpha}).$$

On Γ_+ ,

$$\nabla \log \rho(x) = \sum_{\alpha \in R_+} k(\alpha) \frac{\alpha}{\langle \alpha, x \rangle},$$

and therefore

$$\rho \partial_{d_{g,j}} \rho^{-1} = \partial_{d_{g,j}} - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle}.$$

Conjugation by V cancels the diagonal terms and gives (3).

Similarly, pairing α and $-\alpha$ in the formula for the Dunkl Laplacian and using $\|\alpha\|^2 = 2$, we have

$$\begin{aligned} (U(-\Delta_{\text{Dunkl}})U^{-1}h)_g &= -\Delta h_g - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{\partial_\alpha h_g}{\langle \alpha, x \rangle} \\ &\quad + 2 \sum_{\alpha \in R_+} k(\alpha) \frac{h_g - h_{g\sigma_\alpha}}{\langle \alpha, x \rangle^2}. \end{aligned}$$

Writing $\phi = \log \rho$,

$$\rho(-\Delta - 2\nabla \phi \cdot \nabla) \rho^{-1} = -\Delta + |\nabla \phi|^2 + \Delta \phi,$$

while

$$\Delta \phi = - \sum_{\alpha \in R_+} k(\alpha) \frac{\|\alpha\|^2}{\langle \alpha, x \rangle^2} = -2 \sum_{\alpha \in R_+} \frac{k(\alpha)}{\langle \alpha, x \rangle^2}.$$

The latter term cancels the diagonal part of

$$2 \sum_{\alpha \in R_+} k(\alpha) \frac{I_{\mathbb{C}^G} - S_\alpha}{\langle \alpha, x \rangle^2},$$

and hence

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2},$$

which proves (4).

Assume now that $N > 1$, fix $\chi \in C_c^\infty(\Gamma_+)$, and set

$$K := \text{supp } \chi.$$

On Γ_+ , write

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + W(x), \quad W(x) = |\nabla \log \rho(x)|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Thus W is a smooth Hermitian matrix-valued function on Γ_+ .

Choose open sets

$$K \Subset \Omega_0 \Subset \Omega_1 \Subset \Gamma_+$$

and $\eta \in C_c^\infty(\Gamma_+)$ such that

$$0 \leq \eta \leq 1, \quad \eta = 1 \quad \text{on a neighborhood of } \overline{\Omega_1}.$$

Choose a flat torus M containing $\text{supp } \eta$ in one Euclidean coordinate chart. Since η is compactly supported in that chart, the differential operator $M_\eta \tilde{P} M_\eta$ extends smoothly by zero to M . Define on $L^2(M; \mathbb{C}^G)$

$$(6) \quad P_0 := M_\eta \tilde{P} M_\eta + M_{1-\eta}(-\Delta_M + 1)M_{1-\eta}.$$

It is a nonnegative self-adjoint elliptic differential operator of order 2. Indeed,

$$\langle P_0 u, u \rangle = \langle \tilde{P}(\eta u), \eta u \rangle + \|\nabla((1-\eta)u)\|_2^2 + \|(1-\eta)u\|_2^2 \geq 0,$$

and its principal symbol is

$$p_{0,2}(x, \xi) = (\eta(x)^2 + (1-\eta(x))^2)|\xi|^2 I_{\mathbb{C}^G},$$

which is elliptic since

$$\eta^2 + (1-\eta)^2 \geq \frac{1}{2}.$$

Moreover,

$$(7) \quad P_0 = \tilde{P} \quad \text{on } \Omega_1.$$

Let

$$H_t := e^{-t\tilde{P}}, \quad H_t^0 := e^{-tP_0}.$$

We first consider

$$B_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} H_t^0 dt.$$

Let Π_0 be the orthogonal projection onto $\ker P_0$. Since P_0 is elliptic on the closed manifold M , Π_0 is smoothing. Applying the complex powers calculus [?, Theorem 2 and p. 290] to $P_0 + \Pi_0$, we obtain

$$(P_0 + \Pi_0)^{-1/2} \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G)$$

with principal symbol $p_{0,2}^{-1/2}$.

Set

$$F(\lambda) := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} e^{-t\lambda} dt, \quad \lambda \geq 0.$$

Then $B_0 = F(P_0)$, and for $\lambda > 0$,

$$F(\lambda) - \lambda^{-1/2} = -\frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} e^{-t\lambda} dt.$$

Hence

$$F(\lambda) - \lambda^{-1/2} = O(\lambda^{-m}) \quad (\lambda \rightarrow \infty)$$

for every $m > 0$. On $\ker P_0$ the difference between B_0 and $(P_0 + \Pi_0)^{-1/2}$ is finite rank. It follows from elliptic spectral regularity that

$$B_0 - (P_0 + \Pi_0)^{-1/2} \in \Psi^{-\infty}(M; \mathbb{C}^G).$$

Consequently,

$$(8) \quad B_0 \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G), \quad \sigma_{-1}(B_0)(x, \xi) = p_{0,2}(x, \xi)^{-1/2}.$$

In particular, by (7),

$$(9) \quad \sigma_{-1}(B_0)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}, \quad x \in \Omega_1.$$

We next compare H_t and H_t^0 in the interior. Choose $\zeta \in C_c^\infty(\Omega_1)$ such that

$$\zeta = 1 \quad \text{on a neighborhood of } K.$$

Since $P_0 = \tilde{P}$ on Ω_1 , Duhamel's formula gives

$$(10) \quad M_\chi(H_t - H_t^0)M_\chi = - \int_0^t M_\chi H_{t-s} [\tilde{P}, M_\zeta] H_s^0 M_\chi ds.$$

Indeed, for data supported in K ,

$$v(t) := M_\zeta H_t^0 M_\chi u$$

satisfies

$$(\partial_t + \tilde{P})v = [\tilde{P}, M_\zeta] H_t^0 M_\chi u, \quad v(0) = M_\chi u,$$

which yields (10).

Let $h_t(x, y)$ be the Dunkl heat kernel. The integral kernel of H_t , with respect to Lebesgue measure on Γ , is the matrix

$$(11) \quad \tilde{h}_t(x, y)_{g, \ell} = \rho(x)\rho(y) h_t(gx, \ell y), \quad g, \ell \in G.$$

For $x, y \in \Gamma$,

$$d(gx, \ell y) = d(x, y) = |x - y|.$$

Together with the mixed derivative estimate for the Dunkl heat kernel [?, Theorem 3.1], this implies that, whenever $E, F \Subset \Gamma_+$ and $\text{dist}(E, F) > 0$, for every pair of multi-indices α, β and every $m > 0$,

$$(12) \quad \sup_{\substack{x \in E \\ y \in F}} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

The heat kernel of the smooth elliptic operator P_0 satisfies the same off-diagonal estimate on M .

Since

$$\text{supp}[\tilde{P}, M_\zeta] \cap K = \emptyset,$$

the two heat kernels occurring in (10) are evaluated between compact sets separated by a positive distance. The standard Gaussian bounds underlying (12) therefore give, for every multi-indices α, β and every $m > 0$,

$$(13) \quad \sup_{x, y} \left\| \partial_x^\alpha \partial_y^\beta K_{M_\chi(H_t - H_t^0)M_\chi}(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

Hence

$$(14) \quad S_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} M_\chi(H_t - H_t^0)M_\chi dt \in \Psi^{-\infty}.$$

It remains to treat the large-time part. Let $K_1 \Subset \Gamma_+$ be a compact set containing K . By (11), the heat-kernel derivative estimate cited above, and (2), for $x, y \in K_1$ and $t \geq 1$,

$$V(gx, \ell y, \sqrt{t})^{-1} \lesssim_{K_1} t^{-N/2}.$$

Since ρ and all its derivatives are bounded on K_1 , it follows that, for every pair of multi-indices α, β ,

$$(15) \quad \sup_{x, y \in K_1} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| \lesssim_{\alpha, \beta, K_1} t^{-N/2}, \quad t \geq 1.$$

Since $N > 1$,

$$\int_1^\infty t^{-1/2} t^{-N/2} dt < \infty.$$

Thus all derivatives of

$$K_{S_\infty}(x, y) := \frac{\chi(x)\chi(y)}{\sqrt{\pi}} \int_1^\infty t^{-1/2} \tilde{h}_t(x, y) dt$$

converge locally uniformly, and therefore

$$(16) \quad S_\infty := \frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} M_\chi H_t M_\chi dt \in \Psi^{-\infty}.$$

For completeness, the preceding integrals give the canonical bounded localization of $\tilde{P}^{-1/2}$. Indeed, for $\varepsilon > 0$,

$$(\tilde{P} + \varepsilon)^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\varepsilon t} H_t dt$$

by the spectral theorem. The estimates (13) and (15) imply that

$$M_\chi(\tilde{P} + \varepsilon)^{-1/2} M_\chi$$

converges in operator norm, as $\varepsilon \downarrow 0$, to

$$\frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} M_\chi H_t M_\chi dt.$$

On the natural domain of $\tilde{P}^{-1/2}$, this limit agrees with $M_\chi \tilde{P}^{-1/2} M_\chi$; hence it is its bounded extension.

Combining (14), (16), and the definition of B_0 , we obtain

$$M_\chi \tilde{Q} M_\chi = M_\chi B_0 M_\chi + S_0 + S_\infty.$$

By (8), $M_\chi B_0 M_\chi$ is a classical matrix-valued pseudodifferential operator of order -1 , while $S_0 + S_\infty$ is smoothing. Therefore

$$M_\chi \tilde{Q} M_\chi \in \Psi_{\text{cl}}^{-1},$$

and, by (9),

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

If $\chi = 1$ in a neighborhood of x , this reduces to

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G},$$

which proves (5). □

Lemma 6.4. *Let $j \in \{1, \dots, N\}$ and $f \in C_c^\infty(\mathbb{R}^N)$, and set*

$$A_{f,j} := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

Then, for every $\chi \in C_c^\infty(\Gamma_+)$,

$$M_\chi A_{f,j} M_\chi$$

is a compactly supported classical matrix valued pseudodifferential operator of order -1 , with principal symbol

$$(17) \quad \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) = \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Proof. Set

$$F(x) := \text{diag}_{g \in G} (f_g(x)).$$

Since $VUM_fU^{-1}V^{-1} = M_F$ and $R_{\text{Dunkl},j} = T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$,

$$(18) \quad A_{f,j} = [\tilde{T}_j, M_F] \tilde{Q} + \tilde{T}_j [\tilde{Q}, M_F].$$

By (3),

$$(19) \quad \begin{aligned} [\tilde{T}_j, M_F] &= \text{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g \rangle) \\ &+ \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g - f_{g\sigma_\alpha}) \right) S_\alpha. \end{aligned}$$

Fix $\chi \in C_c^\infty(\Gamma_+)$. Choose $\eta, \psi \in C_c^\infty(\Gamma_+)$ such that $\eta = 1$ on a neighborhood of $\text{supp } \chi$ and $\psi = 1$ on a neighborhood of $\text{supp } \eta$, and set

$$Q_\psi := M_\psi \tilde{Q} M_\psi.$$

By Lemma 6.3, Q_ψ is a classical matrix valued pseudodifferential operator of order -1 , and on a neighborhood of $\text{supp } \eta$,

$$(20) \quad q_{-1}(x, \xi) = \|\xi\|^{-1} I_{\mathbb{C}^G}.$$

We use the Kohn–Nirenberg convention

$$\text{Op}(a)u(x) = (2\pi)^{-N} \int_{\mathbb{R}^{2N}} e^{i(x-y) \cdot \xi} a(x, \xi) u(y) dy d\xi.$$

If a and b are classical symbols, the composition formula [?, Theorem 18.1.8, p. 71] gives

$$(21) \quad a \# b \sim \sum_{\beta \in \mathbb{N}_0^N} \frac{1}{\beta!} \partial_\xi^\beta a D_x^\beta b, \quad D_x := \frac{1}{i} \partial_x.$$

Write, on a neighborhood of $\text{supp } \eta$,

$$\sigma(Q_\psi)(x, \xi) \sim q_{-1}(x, \xi) + q_{-2}(x, \xi) + \cdots.$$

By (4),

$$\sigma(\tilde{P})(x, \xi) \sim p_2(x, \xi) + p_1(x, \xi) + p_0(x, \xi),$$

where

$$p_2(x, \xi) = \|\xi\|^2 I_{\mathbb{C}^G}, \quad p_1(x, \xi) = 0.$$

Since $\tilde{Q} = \tilde{P}^{-1/2}$, $\tilde{Q}\tilde{P}\tilde{Q} = I$. Applying (21) on the region where $\psi = 1$ and comparing the homogeneous terms of degree -1 , the terms containing x -derivatives of p_2 and q_{-1} vanish, and hence

$$q_{-2}p_2q_{-1} + q_{-1}p_2q_{-2} = 0.$$

Together with (20), this gives

$$(22) \quad q_{-2}(x, \xi) = 0$$

on a neighborhood of $\text{supp } \eta$.

Applying (21) to the commutator $[Q_\psi, M_F]$, and using (20)–(22), we obtain, on the same neighborhood,

$$\begin{aligned} \sigma_{-2}([Q_\psi, M_F])(x, \xi) &= \frac{1}{i} \sum_{\ell=1}^N \partial_{\xi_\ell} (\|\xi\|^{-1}) \partial_{x_\ell} F(x) \\ (23) \qquad \qquad \qquad &= i \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Since $\psi = 1$ on a neighborhood of $\operatorname{supp} \eta$ and $\eta = 1$ on a neighborhood of $\operatorname{supp} \chi$,

$$\begin{aligned} M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi &= M_\chi[\tilde{T}_j, M_F] Q_\psi M_\chi, \\ M_\chi \tilde{T}_j [\tilde{Q}, M_F] M_\chi &= M_\chi \tilde{T}_j M_\eta [Q_\psi, M_F] M_\chi. \end{aligned}$$

Both operators on the right are classical matrix valued pseudodifferential operators of order -1 .

By (19), (20), and the order-zero part of the composition formula,

$$\begin{aligned} \sigma_{-1}(M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g(x) \rangle) \\ (24) \qquad \qquad \qquad &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

Under the above Kohn–Nirenberg convention, (3) gives

$$\sigma_1(\tilde{T}_j)(x, \xi) = i \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle).$$

Therefore, by (23),

$$\begin{aligned} \sigma_{-1}(M_\chi \tilde{T}_j [\tilde{Q}, M_F] M_\chi)(x, \xi) \\ (25) \qquad \qquad \qquad = -\chi(x)^2 \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle \langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Combining (24) and (25), and setting

$$\theta := \frac{\xi}{\|\xi\|},$$

we obtain

$$\begin{aligned} \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha \\ &= \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}(x, \theta). \end{aligned}$$

This proves (17). $\square$

Lemma 6.5. *Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Then*

$$\left(\int_\Gamma \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. By the definition of $\mathbf{c}_{f,j}$, the unitary invariance of the Schatten norm, and $\|d_{g,j}\| = 1$,

$$\begin{aligned} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N} &\lesssim_{R,k} \left(\sum_{g \in G} |\nabla f_g(x)|^N \right)^{1/N} \\ &\quad + \sum_{\alpha \in R_+} \left(\sum_{g \in G} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N \right)^{1/N}. \end{aligned}$$

The chamber decomposition used in Lemma 6.3 and the orthogonality of $g \in G$ imply

$$\sum_{g \in G} \int_{\Gamma} |\nabla f_g(x)|^N dx = \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N.$$

Moreover, since $\|\alpha\|^2 = 2$,

$$|x - \sigma_\alpha x| = \sqrt{2} |\langle \alpha, x \rangle|,$$

and hence

$$\left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right| = \sqrt{2} \mathcal{G}_{g, g\sigma_\alpha} f(x)$$

for almost every $x \in \mathbb{R}^N$. By Lemma ??,

$$\int_{\Gamma} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N dx \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N$$

uniformly in $g \in G$ and $\alpha \in R_+$. Since G and R_+ are finite,

$$(26) \quad \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \lesssim_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

For the converse estimate, let P_g denote the orthogonal projection of $\mathbb{C}^G$ onto the g -th coordinate and define

$$\mathcal{E}(A) := \sum_{g \in G} P_g A P_g, \quad A \in \mathcal{L}(\mathbb{C}^G).$$

Thus, $\mathcal{E}(A)$ is the diagonal part of A . The pinching inequality for unitarily invariant norms [?, p. 97, (IV.51)–(IV.52)] gives

$$(27) \quad \|\mathcal{E}(A)\|_{S^N} \leq \|A\|_{S^N}.$$

For every $\alpha \in R_+$, S_α has zero diagonal, since $g\sigma_\alpha \neq g$ for all $g \in G$. Hence, by the definition of $\mathbf{c}_{f,j}$,

$$\mathcal{E}(\mathbf{c}_{f,j}(x, \theta)) = \text{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right).$$

Therefore, by (27),

$$(28) \quad \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N \geq \sum_{g \in G} \left| \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right|^N.$$

For each $g \in G$, set

$$y := gx, \quad \eta := g\theta.$$

Since

$$\nabla f_g(x) = g^{-1} \nabla f(y), \quad d_{g,j} = g^{-1} e_j,$$

and g is orthogonal,

$$\begin{aligned} \langle d_{g,j}, \nabla f_g(x) \rangle &= \partial_j f(y), \\ \langle d_{g,j}, \theta \rangle &= \eta_j, \\ \langle \theta, \nabla f_g(x) \rangle &= \langle \eta, \nabla f(y) \rangle. \end{aligned}$$

Using the chamber decomposition and the orthogonal invariance of surface measure, (28) therefore gives

$$\begin{aligned} & \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N d\theta dx \\ (29) \quad & \geq \int_{\mathbb{R}^N} \int_{\mathbb{S}^{N-1}} |\partial_j f(y) - \eta_j \langle \eta, \nabla f(y) \rangle|^N d\eta dy. \end{aligned}$$

For $v \in \mathbb{C}^N$, set

$$\Phi_j(v) := \left(\int_{\mathbb{S}^{N-1}} |v_j - \eta_j \langle \eta, v \rangle|^N d\eta \right)^{1/N}.$$

Then

$$(30) \quad \Phi_j(v) \simeq_N \|v\|.$$

Indeed, Φ_j is a seminorm on $\mathbb{C}^N$. If $\Phi_j(v) = 0$, continuity gives

$$v_j - \eta_j \langle \eta, v \rangle = 0, \quad \eta \in \mathbb{S}^{N-1}.$$

Since $N > 1$, there exists $\eta \in \mathbb{S}^{N-1}$ with $\eta_j = 0$, and therefore $v_j = 0$. It follows that

$$\langle \eta, v \rangle = 0$$

whenever $\eta_j \neq 0$. Since such η form a dense subset of $\mathbb{S}^{N-1}$, continuity yields $\langle \eta, v \rangle = 0$ for every $\eta \in \mathbb{S}^{N-1}$, and hence $v = 0$. Thus Φ_j is a norm, and (30) follows from the equivalence of norms on the finite dimensional space $\mathbb{C}^N$.

Applying (30) with $v = \nabla f(y)$ in (29), we obtain

$$\|\nabla f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_N \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Combining this estimate with (26) proves the assertion. $\square$

Lemma 6.6. *Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in C_c^\infty(\mathbb{R}^N)$. Then*

$$(31) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Proof. Define

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

By Lemma ?? (??), there exist $\chi_m \in C_c^\infty(\Gamma_+)$, $0 \leq \chi_m \leq 1$, such that $\chi_m(x) \rightarrow 1$ for every $x \in \Gamma_+$ and

$$(32) \quad \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \rightarrow 0.$$

Set

$$A_{f,m} := M_{\chi_m} A_f M_{\chi_m}, \quad K_m := \text{supp } \chi_m.$$

By Lemma 6.4 (17), $A_{f,m}$ is a compactly supported classical matrix valued pseudodifferential operator of order -1 on Γ_+ , with

$$(33) \quad \sigma_{-1}(A_{f,m})(x, \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left(x, \frac{\xi}{\|\xi\|} \right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Choose a bounded open set U_m with $K_m \Subset U_m \Subset \Gamma_+$, and choose $L_m > 0$ such that

$$L_m > \sup_{x,y \in U_m} \|x - y\|_\infty.$$

For

$$\mathbb{T}_m^N := \mathbb{R}^N / L_m \mathbb{Z}^N,$$

the quotient map restricts to a diffeomorphism

$$\kappa_m : U_m \rightarrow \Omega_m := \kappa_m(U_m),$$

which is a local isometry. Since the Schwartz kernel of $A_{f,m}$ is supported in $K_m \times K_m$, with respect to

$$L^2(\Gamma; \mathbb{C}^G) = L^2(U_m; \mathbb{C}^G) \oplus L^2(\Gamma \setminus U_m; \mathbb{C}^G)$$

we have

$$A_{f,m} = B_{f,m} \oplus 0$$

for some $B_{f,m}$ on $L^2(U_m; \mathbb{C}^G)$.

Let

$$W_m : L^2(U_m; \mathbb{C}^G) \rightarrow L^2(\Omega_m; \mathbb{C}^G), \quad W_m u := u \circ \kappa_m^{-1}.$$

Then W_m is unitary. By the invariance of classical pseudodifferential operators under smooth coordinate changes [?, Theorem 18.1.17],

$$T_{f,m} := W_m B_{f,m} W_m^{-1} \in \Psi_{\text{cl}}^{-1}(\Omega_m; \mathbb{C}^G).$$

Its Schwartz kernel is supported in $\kappa_m(K_m) \times \kappa_m(K_m) \Subset \Omega_m \times \Omega_m$. Choose $\vartheta_m \in C_c^\infty(\Omega_m)$ equal to 1 on a neighborhood of $\kappa_m(K_m)$. Then

$$T_{f,m} = M_{\vartheta_m} T_{f,m} M_{\vartheta_m}.$$

Hence the zero extension of the Schwartz kernel of $T_{f,m}$ from $\Omega_m \times \Omega_m$ to $\mathbb{T}_m^N \times \mathbb{T}_m^N$ is a classical pseudodifferential kernel of order -1 : locally in $\Omega_m \times \Omega_m$ this follows from $T_{f,m} \in \Psi_{\text{cl}}^{-1}$, while near the boundary of $\Omega_m \times \Omega_m$ the kernel vanishes. Denote the corresponding operator on $\mathbb{T}_m^N$ by $\hat{A}_{f,m}$. Thus,

$$(34) \quad \hat{A}_{f,m} \in \Psi_{\text{cl}}^{-1}(\mathbb{T}_m^N; \mathbb{C}^G).$$

Since κ_m is a local isometry, by (33) and the transformation law for principal symbols,

$$(35) \quad \sigma_{-1}(\hat{A}_{f,m})(\kappa_m(x), \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left(x, \frac{\xi}{\|\xi\|} \right), \quad x \in U_m, \quad \xi \neq 0,$$

and the principal symbol vanishes outside Ω_m .

With respect to

$$L^2(\mathbb{T}_m^N; \mathbb{C}^G) = L^2(\Omega_m; \mathbb{C}^G) \oplus L^2(\mathbb{T}_m^N \setminus \Omega_m; \mathbb{C}^G),$$

we have

$$\widehat{A}_{f,m} = T_{f,m} \oplus 0.$$

Since $T_{f,m}$ and $B_{f,m}$ are unitarily equivalent,

$$(36) \quad \mu(t, \widehat{A}_{f,m}) = \mu(t, A_{f,m}), \quad t > 0.$$

By (34) and the Weyl law [?, Theorem 6.1],

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, \widehat{A}_{f,m}) = \left(\frac{1}{N(2\pi)^N} \int_{S^* \mathbb{T}_m^N} \text{tr}_{\mathbb{C}^G} |\sigma_{-1}(\widehat{A}_{f,m})(z, \theta)|^N dz d\theta \right)^{1/N}.$$

Since the metric on $\mathbb{T}_m^N$ is flat, combining (35) with (36), we obtain

$$(37) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_{f,m}) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N},$$

where χ_m is identified with its zero extension to Γ .

By Lemma 6.5,

$$(x, \theta) \mapsto \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N$$

belongs to $L^1(\Gamma \times \mathbb{S}^{N-1})$. Since $0 \leq \chi_m \leq 1$, $\chi_m \rightarrow 1$ on Γ_+ , and $\Gamma \setminus \Gamma_+$ is contained in finitely many hyperplanes, the dominated convergence theorem implies

$$(38) \quad \begin{aligned} & \lim_{m \rightarrow \infty} \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \\ &= \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}. \end{aligned}$$

By (32), $A_{f,m} \rightarrow A_f$ in $S^{N,\infty}$. Combining (37), (38), and Lemma 6.7, we obtain

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_f) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Since U and V are unitary,

$$\mu(t, A_f) = \mu(t, [R_{\text{Dunkl},j}, M_f]), \quad t > 0.$$

This completes the proof of Lemma 6.6. $\square$

Lemma 6.7. *Let $0 < p < \infty$, let $\mathcal{H}$ be a Hilbert space, and let $A, A_m \in S^{p,\infty}(\mathcal{H})$, $m \geq 1$, satisfy*

$$\|A_m - A\|_{S^{p,\infty}} \longrightarrow 0.$$

Suppose that, for every $m \geq 1$, the limit

$$c_m := \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A_m)$$

exists. Then both limits

$$\lim_{m \rightarrow \infty} c_m \quad \text{and} \quad \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A)$$

exist and

$$\lim_{t \rightarrow \infty} t^{1/p} \mu(t, A) = \lim_{m \rightarrow \infty} c_m.$$

Proof. The assertion is the Birman–Solomyak approximation lemma; see [?, Section 11.6]. $\square$

Lemma 6.8. *Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Then there exists $f_m \in C_c^\infty(\mathbb{R}^N)$ such that*

$$\begin{aligned} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} &\longrightarrow 0, \\ \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &\longrightarrow 0, \\ \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &\longrightarrow 0. \end{aligned}$$

Proof. By the density of $C_c^\infty(\mathbb{R}^N)$ in $\dot{W}^{1,N}(\mathbb{R}^N)$, choose $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$\|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0.$$

By Theorem ?? and Lemma ??,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &= \|[R_{\text{Dunkl},j}, M_{f_m-f}]\|_{S^{N,\infty}} \\ &\lesssim \|f_m - f\|_{OSC^{N,\infty}(\mathbb{R}^N)} \lesssim \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

By the linearity of $f \mapsto \mathfrak{c}_{f,j}$ and Lemma 6.5,

$$\begin{aligned} \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &= \|\mathfrak{c}_{f_m-f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \\ &\lesssim_{N,R,k} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

This completes the proof of Lemma 6.8. $\square$

Theorem 6.9. *Assume that $N > 1$ and $j \in \{1, \dots, N\}$. Then, for every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$,*

(39)

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Moreover,

$$(40) \quad \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. By Lemma 6.8, choose $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$(41) \quad \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} \longrightarrow 0,$$

$$(42) \quad \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow 0.$$

By Lemma 6.6, for every m ,

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_{f_m}]) = \frac{1}{(N(2\pi)^N)^{1/N}} \|\mathfrak{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

By (42),

$$\|\mathfrak{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow \|\mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

Combining this convergence with (41), we apply Lemma 6.7 with $p = N$ and obtain (39). Finally, (40) follows from Lemma 6.5. This completes the proof of Theorem 6.9. $\square$

APPENDIX A. ZHIJIE LEMMA

Lemma A.1. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. Then there exists a constant $C > 0$ such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in \mathcal{E}} Q$ we may choose a cube $Q_x \in \mathcal{E}$ of minimal volume such that $x \in Q_x$. Every $Q \in \mathcal{E}$ containing x is then a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix $t > 0$ and let $\mathcal{E}$ be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since $\mathcal{E}$ is arbitrary,

$$\#\left\{ Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t \right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak ℓ^s estimate follows. This completes the proof of Lemma A.1. $\square$

REFERENCES

- [1] J. Anker, J. Dziubański, and A. Hejna. Harmonic functions, conjugate harmonic functions and the Hardy space H^1 in the rational Dunkl setting. *J. Fourier Anal. Appl.*, 25(5):2356–2418, 2019.
- [2] Nicolas Bourbaki. *Lie groups and Lie algebras. Chapters 4–6*. Elements of Mathematics (Berlin). Springer-Verlag, Berlin, 2002. Translated from the 1968 French original by Andrew Pressley. [2](#)
- [3] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [3](#)
- [4] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [3](#)
- [5] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at <https://arxiv.org/abs/2403.08249>, 2024. [3](#)
- [6] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [16](#)
- [7] G. Heckman. Coxeter groups. Lecture notes available at <https://www.math.ru.nl/~heckman/CoxeterGroups.pdf>. Radboud University Nijmegen, 2018. [19](#)
- [8] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [2](#)
- [9] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [3](#)
- [10] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [3](#)
- [11] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [3](#)
- [12] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon-Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. [3](#)
- [13] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. [3](#)
- [14] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. [3](#)

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